Fiscal Year:	FY 2024	**Task Last Updated:**	FY 03/03/2024
PI Name:	Kim, Jungho Ph.D.		
Project Title:	Gravity Effects on Flow Boiling Heat Transfer Using Local Heat Transfer Distribution Methods		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20742-0001	**Congressional District:**	5
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2015	**End Date:**	05/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2022 per J. McQuillen/GRC (Ed., 3/30/22) NOTE: End date changed to 5/31/2021 per NSSC information (Ed., 8/12/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AK16G		
Performance Goal No.:			
Performance Goal Text:			
	EDITOR'S NOTE (5/17/2016): Continuation of "Measurement and Analysis of Gravity Effects on Flow Boiling Heat Transfer Using IR Thermometry" grant # NNX12AF31A, with same Principal Investigator, Dr. Jungho Kim. Added to Task Book when received information. This effort will be conducted with the European Space Agency (ESA) in one of the facilities/test articles that they are developing. Fluorescent paints that are sensitive to oxygen concentration (pressure sensitive paints, PSP) and temperature (temperature sensitive paints,TSP) have been used to measure shear stress, surface pressure, and wall temperature distributions in aerodynamic and fluid mechanics applications since the 1980s. It has seen very limited use for		

Task Description:	phase-change applications, however. If only temperature is to be measured, the paint is encapsulated within a binder or between impermeable barriers to eliminate sensitivity to oxygen. The molecules are typically stimulated using blue-UV light, and emission at a longer wavelength occurs. The intensity of the emitted light decreases with temperature and oxygen concentration, and a filter is used to remove the excitation light before the emitted light is measured by a camera. The emission lifetime is typically on the order of microseconds, much faster than the boiling processes being measured, so the system frequency response is usually limited by the thermal response of the materials onto which the paint is bonded. The advantages of TSPs over infrared thermography and other techniques such as microheater arrays or microbolometers to measure temperature are numerous. Readily available CCD or CMOS cameras can be used to measure the intensity distribution instead of expensive IR (infrared) cameras. Since TSPs respond very quickly, the transient response of TSP based measurements is limited in practice only by the speed of the camera, which is typically much faster than IR cameras. Also, since visible light is used for illumination and sensing, glass, clear plastics, PDMS (polydimethylsiloxane) and other commonly used materials can be used in place of expensive and difficult to machine IR transparent materials (e.g., silicon, sapphire, CaF2) as substrates. Calibration of TSPs can be performed without having to consider emission from the transparent substrate they are placed on, unlike for IR measurements. Finally, the wavelength of the light used (~0.5 mm for TSPs) is much smaller than the 3-5 µm or 8-12 µm light IR cameras are sensitive to, allowing much higher spatial resolution measurements. The diffraction limit for 0.6 µm light in air is ~0.3 µm. TSPs have quite high sensitivity (approximately -2%/°C), and TSPs are available that can be used over a wide temperature range. Although TSPs have been used for aerospace applications for many years, very little work has been performed using this technique for two-phase flows. We have developed methods whereby temperature and heat flux distributions can be measured during pool boiling and flow boiling so models of pool and flow boiling can be verified. A test apparatus has been developed and this technique will be used to measure gravity effects on flow boiling during a series of low-g aircraft flights in March, 2017.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data is being used to develop and validate sub-models of the flow within each flow regime. Understanding of the heat transfer mechanisms will enable space-based heat exchangers to be designed with much more confidence. Although this work is primarily aimed at thermal management subsystems and development of lightweight evaporators, it also impacts other systems for novel power and transmission, advanced chemical propulsion, and energy storage.
Task Progress:	Flow boiling is a method of phase change heat transfer used widely in electronics cooling, refrigeration, air conditioning, and other areas where stable temperatures are needed. An area of interest is spaceflight systems, where efficient heat transfer is desired to minimize mass, power requirements, and cost. When compared to terrestrial gravity conditions, the heat transfer of flow boiling in microgravity typically depreciates. This depreciation has been documented across multiple experimental studies performed by teams using different fluids, tube geometries, and flow regimes over the past three decades. Though select experimental microgravity flow boiling heat transfer data are available in the literature, holistic results are sparse due to the cost and limited availability of microgravity research. The two-phase heat transfer mechanisms responsible for the depreciation are therefore not well known, and so heat transfer models for variable gravity flow boiling do not exist. The goal of the proposed study is to develop models for flow boiling heat transfer through a tube as a function of gravity by identifying the effect of gravity on different heat transfer mechanisms. The scope of this proposal involves modeling three microgravity flow regimes (bubbly, slug, and annular flow) to serve as baseline predictions for flow boiling heat transfer without the influence of gravity. Additional gravity effects can be identified using partial and hyper-gravity data. Experiments have been performed aboard parabolic flights and on the ground at various flow rates, heating rates, and inlet subcoolings in microgravity, hyper-gravity, Lunar gravity, Martian gravity, and terrestrial gravity. Results from the experiments showed that negligible slip velocity plays an important role in modeling flow boiling heat transfer. Simulations using modified single-phase models of an accelerating flow were performed which predicted microgravity flow boiling heat transfer well in the nucleate boiling regime. Additional experiments concerning terrestrial gravity quenching heat transfer have been performed to address research gaps in microgravity cryogen chilldown studies. Quenching heat transfer coefficients were recorded in the nucleate boiling regime and compared with correlations. The correlations were able to predict heat transfer for room temperature fluids much more accurately than for cryogenic fluids. Scaling parameters must be tuned to match cryogen data to examine the large disparity between cryogenic quenching heat transfer data and correlations observed in the literature.
Bibliography Type:	Description: (Last Updated: 03/13/2026)
Articles in Peer-reviewed Journals	Lebon MT, Hammer CF, Kim J. "Using a modified single-phase model to predict microgravity flow boiling heat transfer in the bubbly flow regime." Experimental Heat Transfer. 2021 Jul 29;34(5):474-92. https://doi.org/10.1080/08916152.2020.1782534 , Jul-2021